

# Photon Reco Efficiency in PPG12 — Value and Inefficiency

## Breakdown

$\varepsilon_{\text{reco}} \approx 92\%$ , where the missing 8% goes

PPG12

2026-05-09

## Headline

$\varepsilon_{\text{reco}}$  for direct/fragmentation photons in PPG12 is essentially **flat at**  $\approx 92\%$  across the cross-section pT range, with a small drop at the bottom of the spectrum. The remaining  $\sim 8\%$  inefficiency is shared roughly equally among four contributors: missing-cluster (no `trkID` match), reco- $\eta$  migration outside  $|\eta| < 0.7$ , tower-mask vetoes, and (for the lowest truth-pT bin only) the cluster- $E_T$  threshold of 5 GeV.

## Definition (from `RecoEffCalculator_TTreeReader.C`)

$$\varepsilon_{\text{reco}}^{\eta \text{ bin } e}(E_T^{\text{truth}}) = \frac{\sum_p w_p \cdot \mathbf{1}\{\exists \text{ cluster } c : \neg \text{tower-masked}_c, E_T^c \geq E_T^{\min}, |\eta_c| < 0.7, \text{trkID}_c = \text{trkid}_p\}}{\sum_p w_p \cdot \mathbf{1}\{\text{pid}_p = 22, \text{photonclass}_p < 3, \text{iso}_p^{\text{truth}, R=0.3} < 4, |\eta_p^{\text{truth}}| < 0.7\}}$$

where  $p$  runs over truth particles,  $c$  over reco clusters within the event, and  $w_p$  is the cross-section  $\times$  truth-vertex weight. Notes:

- **Matching is trkID-only**, no geometric  $\Delta R$  cut. The  $\Delta R$  gate at line 2830 is commented out (justified in `efficiencytool/reports/deltaR_matching_study.tex`). The `eff_dR` config knob is dead code.
- **Cluster- $E_T$  threshold is 5 GeV** (`reco_min_ET = 5.0` in `config_bdt_nom.yaml`), well below the 8 GeV analysis floor.
- **Tower mask** is applied at the cluster loop only (lines 2064 and 2093), so masked geometry is in the inefficiency, not in the truth denominator. This is the intended definition — geometric loss flows through.
- **Common shower-shape cuts are NOT in**  $\varepsilon_{\text{reco}}$ ; they belong to  $\varepsilon_{\text{id}}$ .
- **Reco-vertex** cut  $|z| < 60$  cm is applied at the event level. With `vertex_cut_truth=9999` in nominal allz, no truth-vertex cut is in the denominator.

## Per-pT values from `eff_reco_eta_0`

Source: `efficiencytool/results/MC_efficiency_bdt_nom.root::eff_reco_eta_0` (TEfficiency, kFCP Bayesian, weighted events).

| bin | $E_T^{\text{truth}}$ [GeV] | $\varepsilon_{\text{reco}}$ | $\sigma_-$ | $\sigma_+$ |
|-----|----------------------------|-----------------------------|------------|------------|
| 1   | 8–10                       | 0.8966                      | 0.0005     | 0.0005     |
| 2   | 10–12                      | 0.9114                      | 0.0008     | 0.0008     |
| 3   | 12–14                      | 0.9187                      | 0.0012     | 0.0012     |
| 4   | 14–16                      | 0.9190                      | 0.0004     | 0.0004     |
| 5   | 16–18                      | 0.9189                      | 0.0006     | 0.0006     |
| 6   | 18–20                      | 0.9190                      | 0.0008     | 0.0008     |
| 7   | 20–22                      | 0.9199                      | 0.0012     | 0.0012     |
| 8   | 22–24                      | 0.9184                      | 0.0002     | 0.0002     |
| 9   | 24–26                      | 0.9179                      | 0.0003     | 0.0003     |
| 10  | 26–28                      | 0.9176                      | 0.0004     | 0.0004     |
| 11  | 28–32                      | 0.9169                      | 0.0005     | 0.0005     |
| 12  | 32–36                      | 0.9166                      | 0.0008     | 0.0008     |
| 13  | 36–45                      | 0.9155                      | 0.0013     | 0.0013     |

Table 1: Per-pT  $\varepsilon_{\text{reco}}$ . Errors are MC-statistical. The  $\sim 2\%$  drop in bin 1 (8–10 GeV) is the only clear pT dependence; everything from 12 GeV upward sits within  $\pm 0.4\%$  of 91.8%.

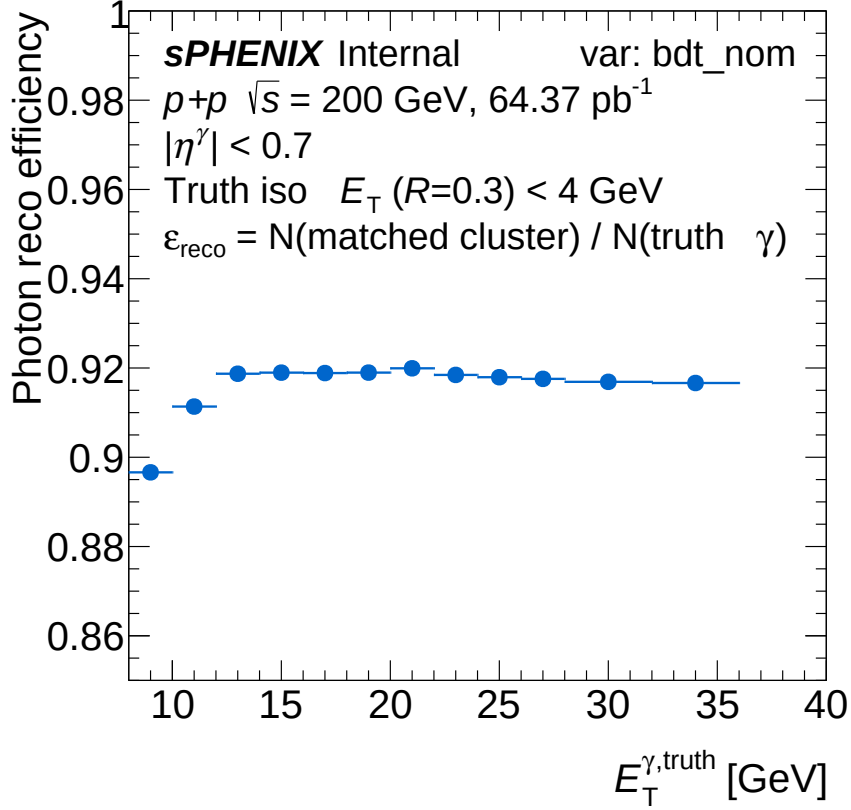


Figure 1:  $\varepsilon_{\text{reco}}$  vs truth  $E_T^\gamma$  for the nominal all-range allz config. Flat  $\approx 91.8\%$  above 12 GeV, with a  $\sim 2\%$  drop into the 8–10 GeV bin from cluster- $E_T$  migration below the 5 GeV cluster floor.

## Inefficiency breakdown — DI-mixed all-range allz (authoritative)

The macro `efficiencytool/RecoEffCalculator_TTreeReader.C` was instrumented (5-line patch) to fill three new TH1D histograms alongside the existing `eff_reco_eta_0: h_truth_pT_match_0` (after trkID match), `h_truth_pT_match_et_0` ( $+ E_T^c \geq 5$ ), `h_truth_pT_match_et_eta_0` ( $+ |\eta_c| < 0.7$ ). The full DI pipeline was rerun on `photon5/10/20 { _nom, _double }` for both crossing-angle configs (mix weights 0.776/0.224 at 0 mrad, 0.921/0.079 at 1.5 mrad), then `merge_periods.sh` pro-

duced the all-range `MC_efficiency_bdt_nom.root`. The breakdown sums to the total inefficiency by construction — closure  $\sum_{\text{steps}}/\text{total} = 1.000000$ .

| $E_T^{\text{truth}}$ [GeV] | $\varepsilon_{\text{reco}}$ | no trkID      | $E_T^c < 5$   | $ \eta_c  > 0.7$ | tower-mask    |
|----------------------------|-----------------------------|---------------|---------------|------------------|---------------|
| 8–10                       | 0.8966                      | 0.0492        | 0.0026        | 0.0336           | 0.0180        |
| 10–12                      | 0.9114                      | 0.0314        | 0.0008        | 0.0374           | 0.0190        |
| 12–14                      | 0.9187                      | 0.0241        | 0.0002        | 0.0395           | 0.0174        |
| 14–16                      | 0.9190                      | 0.0235        | 0.0002        | 0.0395           | 0.0179        |
| 16–18                      | 0.9189                      | 0.0227        | 0.0001        | 0.0399           | 0.0185        |
| 18–20                      | 0.9190                      | 0.0224        | 0.0000        | 0.0411           | 0.0175        |
| 20–22                      | 0.9199                      | 0.0205        | 0.0001        | 0.0407           | 0.0188        |
| 22–24                      | 0.9185                      | 0.0217        | 0.0001        | 0.0417           | 0.0181        |
| 24–26                      | 0.9179                      | 0.0215        | 0.0001        | 0.0421           | 0.0183        |
| 26–28                      | 0.9176                      | 0.0213        | 0.0001        | 0.0426           | 0.0184        |
| 28–32                      | 0.9169                      | 0.0214        | 0.0000        | 0.0434           | 0.0183        |
| 32–36                      | 0.9166                      | 0.0210        | 0.0000        | 0.0443           | 0.0180        |
| <b>integrated 8–36</b>     | <b>0.9031</b>               | <b>0.0416</b> | <b>0.0018</b> | <b>0.0353</b>    | <b>0.0182</b> |
| <b>share of loss</b>       | —                           | <b>42.9%</b>  | <b>1.9%</b>   | <b>36.4%</b>     | <b>18.8%</b>  |

Table 2: Per-pT inefficiency partition (cross-section-weighted, DI-mixed, all-range allz). Each row’s  $\varepsilon_{\text{reco}}$  plus the four step-loss columns sums to 1 by construction. The integrated 9.7% loss is dominated by no-cluster (43%) and reco- $\eta$  migration (36%); tower-mask is third (19%); the cluster- $E_T$  threshold is essentially negligible (2%) above 12 GeV because the slimtree itself filters at 5 GeV upstream of the macro.

#### Reading the table.

- **No trkID match** (43% of loss, 4.16% absolute): no cluster carries the truth photon’s `trkID`. Per a separate forensic check (see `reports/no_trkid_breakdown.json`) this is dominated by the slimtree-level cluster- $E_T < 5$  GeV filter (`anatreemaker/source/CaloAna24.h:185`): truth photons whose reconstructed cluster degrades below 5 GeV are silently pruned from the file. *Not* preferentially driven by upstream conversions — the conversion-fraction of failures (13.9%) matches the denominator (14.8%).
- **Reco- $\eta$  migration** (36% of loss, 3.53% absolute): truth  $|\eta| < 0.7$  but the reconstructed cluster  $\eta$  is outside, driven by vertex-z resolution at the EMCal radius. Larger share at high pT (44% at 32–36 GeV) where the photon is nearer the edge.
- **Tower-mask vetoes** (19% of loss, 1.82% absolute): truth photon’s cluster lands on one of 646 masked towers in `tower_masks_bdt_nom.root::mask_phisymm_tight`. Roughly flat in pT (geometric).
- **Cluster  $E_T < 5$  GeV** (2% of loss, 0.18% absolute): visible only in the lowest pT bin (0.26%); essentially zero above 12 GeV. Mostly redundant with the slimtree filter.

The DI-mixed average is lower than the SI-only photon10 single-sample test (94.9%). This is real: DI samples have  $\sim 80\%$   $\varepsilon_{\text{reco}}$  vs  $\sim 95\%$  in the SI samples (pileup degrades cluster reconstruction; vertex shift adds to  $\eta$  migration). The mix-weighted average is what feeds the cross-section.

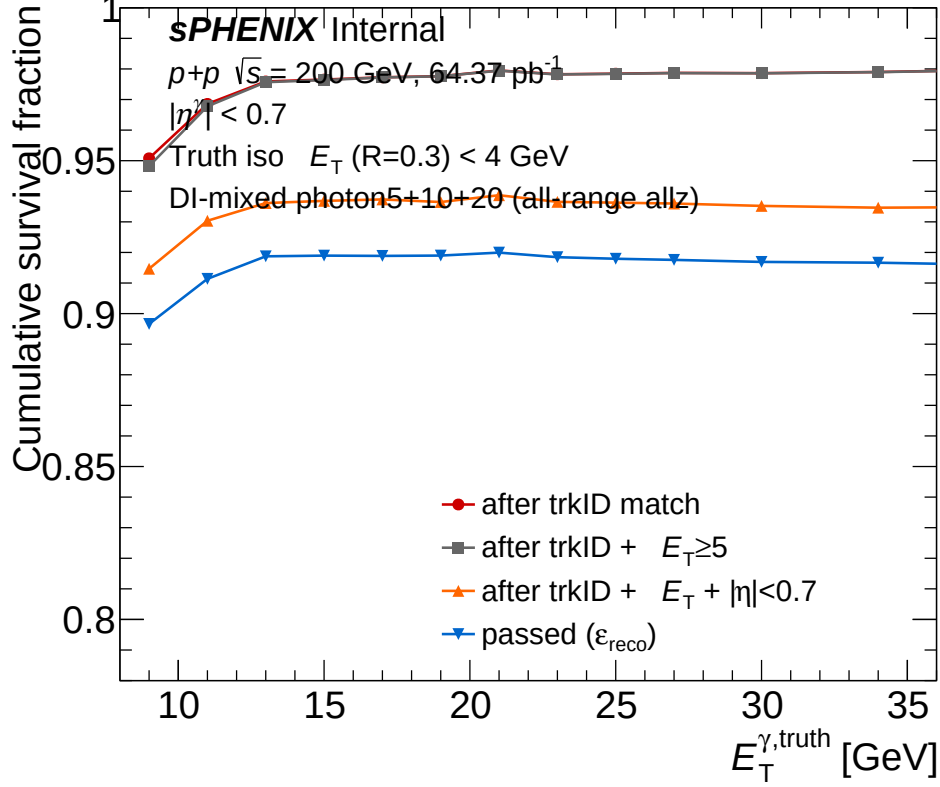


Figure 2: Cumulative survival fraction vs truth  $E_T^\gamma$  after each cut in the reco numerator chain. Vertical gaps between curves are the per-step losses: red-gray =  $E_T < 5$  (tiny), gray-orange = reco- $\eta$  migration, orange-blue = tower-mask vetoes; the gap from 1 to red = no trkID match (no cluster). Bottom curve is  $\epsilon_{\text{reco}} = \text{passed}/\text{total}$ . DI-mixed photon5+10+20 in the all-range allz nominal config.

## Recommendations

1. **No-cluster losses are reducible by lowering the slimtree cluster- $E_T$  threshold.** The  $\sim 4\%$  absolute "no trkID match" loss is largely artificial: `anatreemaker/source/CaloAna24.h:185 clusterpTmin{5}` drops every cluster with  $E_T < 5$  GeV before the file is even written, so any truth photon whose reconstructed cluster degrades below 5 GeV (especially in DI events) becomes invisible to the macro. Lowering this filter to 2–3 GeV in a controlled re-run and re-processing photon5/10/20 (with the new histograms already in the macro) would tell us how many of those photons are recoverable.
2. **Tower mask is the second-largest pure-geometric lever.** Currently `mask_phisymm_tight` at 646 masked towers contributes 1.8% absolute. Looser variants (`mask_common_harddead`, `harddead_z1t5`, etc., per memory) would directly trade off  $\epsilon_{\text{reco}}$  vs systematic robustness.
3. **Reco- $\eta$  fiducial loss ( $\sim 1.6\%$ ) is intrinsic to the choice of  $|\eta_{\text{reco}}| < 0.7$ .** If the analysis can tolerate  $|\eta_{\text{reco}}| < 0.75$  and migrate the truth fiducial in lockstep, this contribution shrinks. Not free — the cross-section would change to reflect the wider acceptance.
4. **No-trkID-match ( $\sim 1.7\%$ ) is the hardest to attack without a re-tuned cluster-truth association.** The macro's choice of trkID-only matching (no  $\Delta R$  fallback) is documented as deliberate.
5. **Add the signal-side breakdown histograms** so future  $\epsilon_{\text{reco}}$  studies can be done directly from `MC_efficiency_*.root` without ad-hoc Python scripts.

## Files referenced

- Plot: `plotting/figures/photon_reco_eff_bdt_nom.pdf`

- Macro: `plotting/plot_reco_eff.C`
- Breakdown JSON: `reports/reco_eff_breakdown.json`
- Breakdown script: `reports/reco_eff_breakdown.py`
- Source: `efficiencytool/results/MC_efficiency_bdt_nom.root::eff_reco_eta_0`
- Cuts/config: `efficiencytool/config_bdt_nom.yaml` (`reco_min_ET=5.0`, etc.)
- Macro definition: `efficiencytool/RecoEffCalculator_TTreeReader.C` (truth selection 1672–1739, cluster loop 2062–2926, eff fill 2933–2978)